

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 1: The Blood Glucose Puzzle: Why Does Glucose Rise and Fall After Eating Ugali?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	C	B	C	B	B	D

USE AFTER: OBSERVE PHASE — Observe: data questions on Wanjiru's table and graph (By how much did glucose rise from 0 to 60 min?)

Q1. Wanjiru fasted, then ate ugali, sukuma wiki and beans. Her blood glucose was 4.5 mmol/L at 0 min and reached its highest value of 8.0 mmol/L at 60 min. By how much did her blood glucose rise from 0 to 60 min?

- A. 8.0 mmol/L
- B. 4.5 mmol/L
- C. 3.5 mmol/L**
- D. 12.5 mmol/L

Correct answer: C. Rise = peak minus starting value = $8.0 - 4.5 = 3.5$ mmol/L. Learners who choose 12.5 added the values instead of finding the difference.

USE AFTER: OBSERVE PHASE — Observe: calculating the rise (3.5 mmol/L, about 78% above the start)

Q2. Wanjiru's blood glucose started at 4.5 mmol/L and rose by 3.5 mmol/L to a peak of 8.0 mmol/L. About what percentage above her starting level was the peak?

- A. About 56%
- B. About 3.5%
- C. About 78%**
- D. About 44%

Correct answer: C. Percentage rise = rise divided by starting value times 100 = $3.5 / 4.5 \times 100 =$ about 78%. Dividing by the peak ($3.5 / 8.0$) gives the common error of about 44%.

USE AFTER: OBSERVE PHASE — Observe: 'Which is faster, rise or fall? Prove it with numbers.'

Q3. Wanjiru's blood glucose rose by 3.5 mmol/L between 0 and 60 min (4.5 to 8.0 mmol/L). It then fell by 3.5 mmol/L between 60 and 180 min (8.0 back to 4.5 mmol/L). What do these numbers show?

- A. The rise and fall cannot be compared because they have different start times.
- B. The rise was faster because the same 3.5 mmol/L change took 60 min going up but 120 min coming down.**
- C. The fall was faster because glucose ended at a lower value than the peak.
- D. The rise and the fall were equally fast because both changes were 3.5 mmol/L.

Correct answer: B. The same change of 3.5 mmol/L took 60 min on the way up and 120 min on the way down, so the rise was about twice as fast. Equal change does not mean equal speed.

USE AFTER: OBSERVE PHASE — Observe: looking at the meal composition card

Q4. A meal card for Wanjiru's lunch lists: ugali about 70 g carbohydrate, beans about 20 g protein and 15 g carbohydrate, sukuma wiki about 3 g protein plus fibre, vitamins and minerals, cooking oil about 10 g fat. Which food in the meal contained the largest amount of carbohydrate?

- A. Sukuma wiki
- B. Cooking oil
- C. Ugali**
- D. Beans

Correct answer: C. Ugali has about 70 g carbohydrate, far more than the 15 g in beans. Sukuma wiki is mostly protein, fibre, vitamins and minerals, and cooking oil is fat.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning organiser and 'What I am sure about / not sure about yet'

Q5. After looking at the graph of blood glucose rising to 8.0 mmol/L and then falling back to 4.5 mmol/L, a group is asked what causes the fall after the peak. Which statement is the most honest scientific position at this stage of the unit?

- A. Glucose is burnt off in the stomach straight after the meal.
- B. The graph shows that glucose falls, but we have not yet explained what causes it, so it is a question to investigate.**
- C. Glucose falls only because the person has stopped eating.
- D. Glucose simply disappears from the blood with time and nothing is involved.

Correct answer: B. The data show the pattern but not its cause. The class still has to investigate it, and saying 'I do not know yet' is acceptable. The other choices are unsupported everyday ideas.

USE AFTER: DQB PHASE — DQB: writing and sorting questions under 'What chemicals are in the food we eat, and what happens to them in our bodies?'

Q6. Learners are writing questions for the Driving Question Board about the glucose graph. Which question is the most specific and can be investigated?

- A. Is glucose good?
- B. What chemicals are in ugali?**
- C. Did Wanjiru enjoy her lunch?
- D. Why is food?

Correct answer: B. 'What chemicals are in ugali?' is clear and can be investigated, for example by testing foods in the lab. The others are vague, opinion-based or not linked to the graph.

USE AFTER: MODEL BUILDING PHASE — Model: initial flow-diagram model (Food chemicals → ? → Glucose in blood → ? → Back to normal)

Q7. Learners draw an initial model of Wanjiru's meal and her glucose graph, using boxes and arrows with question marks where they are unsure. Which skeleton shows the correct order of the parts?

- A. Back to normal → ? → Food chemicals → ? → Glucose in blood
- B. Glucose in blood → ? → Food chemicals → ? → Back to normal
- C. Food chemicals → ? → Back to normal → ? → Glucose in blood
- D. Food chemicals → ? → Glucose in blood (rises to 8.0 mmol/L) → ? → Back to normal (4.5 mmol/L)**

Correct answer: D. The pattern runs from food chemicals, to a rise of glucose in the blood to 8.0 mmol/L, to a return to normal at 4.5 mmol/L. The question marks show the processes still unknown.